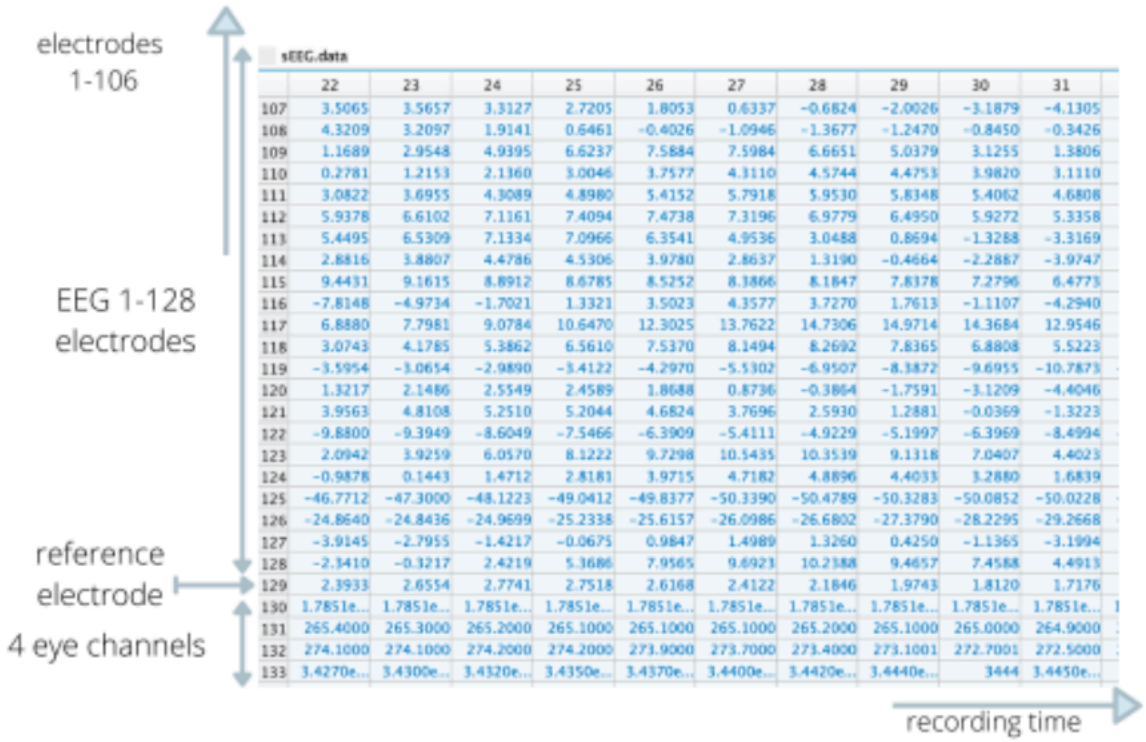
Data Structure Description

sEEG (**s**ynchronised EEG) is a MATLAB structure that contains all the information about the current dataset (synchronized EEG and Eye Tracking)

Below the explanation of the sEEG substructures:

sEEG.data

continuous data matrix :



sEEG.data

	22	23	24	25	26	27	28	29	30	31
107	3.5065	3.5657	3.3127	2.7205	1.8053	0.6337	-0.6824	-2.0026	-3.1879	-4.1305
108	4.3209	3.2097	1.9141	0.6461	-0.4026	-1.0946	-1.3677	-1.2470	-0.8450	-0.3426
109	1.1689	2.9548	4.9395	6.6237	7.5884	7.5984	6.6651	5.0379	3.1255	1.3806
110	0.2781	1.2153	2.1360	3.0046	3.7577	4.3110	4.5744	4.4753	3.9820	3.1110
111	3.0822	3.6955	4.3089	4.8980	5.4152	5.7918	5.9530	5.8348	5.4062	4.6808
112	5.9378	6.6102	7.1161	7.4094	7.4738	7.3196	6.9779	6.4950	5.9272	5.3358
113	5.4495	6.5309	7.1334	7.0966	6.3541	4.9536	3.0488	0.8694	-1.3288	-3.3169
114	2.8816	3.8807	4.4786	4.5306	3.9780	2.8637	1.3190	-0.4664	-2.2887	-3.9747
115	9.4431	9.1615	8.8912	8.6785	8.5252	8.3866	8.1847	7.8378	7.2796	6.4773
116	-7.8148	-4.9734	-1.7021	1.3321	3.5023	4.3577	3.7270	1.7613	-1.1107	-4.2940
117	6.8880	7.7981	9.0784	10.6470	12.3025	13.7622	14.7306	14.9714	14.3684	12.9546
118	3.0743	4.1785	5.3862	6.5610	7.5370	8.1494	8.2692	7.8365	6.8808	5.5223
119	-3.5954	-3.0654	-2.9890	-3.4122	-4.2970	-5.5302	-6.9507	-8.3872	-9.6955	-10.7873
120	1.3217	2.1486	2.5549	2.4589	1.8688	0.8736	-0.3864	-1.7591	-3.1209	-4.4046
121	3.9563	4.8108	5.2510	5.2044	4.6824	3.7696	2.5930	1.2881	-0.0369	-1.3223
122	-9.8800	-9.3949	-8.6049	-7.5466	-6.3909	-5.4111	-4.9229	-5.1997	-6.3969	-8.4994
123	2.0942	3.9259	6.0570	8.1222	9.7298	10.5435	10.3539	9.1318	7.0407	4.4023
124	-0.9878	0.1443	1.4712	2.8181	3.9715	4.7182	4.8896	4.4033	3.2880	1.6839
125	-46.7712	-47.3000	-48.1223	-49.0412	-49.8377	-50.3390	-50.4789	-50.3283	-50.0852	-50.0228
126	-24.8640	-24.8436	-24.9699	-25.2338	-25.6157	-26.0986	-26.6802	-27.3790	-28.2295	-29.2668
127	-3.9145	-2.7955	-1.4217	-0.0675	0.9847	1.4989	1.3260	0.4250	-1.1365	-3.1994
128	-2.3410	-0.3217	2.4219	5.3686	7.9565	9.6923	10.2388	9.4657	7.4588	4.4913
129	2.3933	2.6554	2.7741	2.7518	2.6168	2.4122	2.1846	1.9743	1.8120	1.7176
130	1.7851e...	1.7851e...	1.7851e...	1.7851e...	1.7851e...	1.7851e...	1.7851e...	1.7851e...	1.7851e...	1.7851e...
131	265.4000	265.3000	265.2000	265.1000	265.1000	265.1000	265.2000	265.1000	265.0000	264.9000
132	274.1000	274.1000	274.2000	274.2000	273.9000	273.7000	273.4000	273.1001	272.7001	272.5000
133	3.4270e...	3.4300e...	3.4320e...	3.4350e...	3.4370e...	3.4400e...	3.4420e...	3.4440e...	3444	3.4450e...

sEEG.chanlocs

stores information about the EEG channel locations and channel names.

Here, sEEG.chanlocs is a structure array of length 133 (one record for each of the 128 channels in this dataset + 1 “Cz” central reference electrode + 4 eye channels, as explained below).

The EYE-EEG toolbox parses, imports, and synchronizes simultaneously recorded eye-tracking data and adds it as extra channels to the EEG.

sEEG.event

The EEG structure field contains records of the experimental events that occurred while the data was being recorded, plus the user-defined events.

- Type - contains the event type (e.g. L_saccade, L_fixation, where L corresponds to the left eye)
- latency - contains the event latency in data sample unit
- urevent - contains the index of the event in the original table).
- Duration - the duration of the event
- Endtime - endtime of the event
- Sac_amplitude- for events “L_saccade” and “R_saccade” only. Amplitude of the event (in degrees)
- Sac_endpos_x- for events “L_saccade” and “R_saccade” only.
- Sac_endpos_y - for events “L_saccade” and “R_saccade” only.
- Sac_startpos_x- for events “L_saccade” and “R_saccade” only.
- Sac_startpos_y- for events “L_saccade” and “R_saccade” only.
- Sac_vmax- for events “L_saccade” and “R_saccade” only. Max velocity during execution of the saccade.
- Fix_avgpos_x- average position during the “L_fixation” or “R_fixation” event, coordinate x.
- Fix_avgpos_y-average position during the “L_fixation” or “R_fixation” event, coordinate y.
- Fix_avgpupilsize- average pupil size during the “L_fixation” or “R_fixation” event, coordinate x.

sEEG.event

Fields	type	latency	urevent	duration	endtime	sac_amplitude	sac_endpos_x	sac_endpos_y	sac_startpos_x	sac_startpos_y
1	'12 '	15	1	0	0	0	0	0	0	0
2	'55 '	26	2	0	0	0	0	0	0	0
3	'101 '	31	3	0	0	0	0	0	0	0
4	'L_saccade'	60	282	15	74	3.3000	363	272.5000	265.2000	272
5	'L_fixation'	75	705	76	150	0	0	0	0	0
6	'L_saccade'	151	283	14	164	2.3700	411.1000	329.5000	367.2000	275.8000
7	'L_fixation'	165	706	47	211	0	0	0	0	0
8	'L_saccade'	212	284	72	283	0.5700	418.3000	323.3000	410.6000	338.2000
9	'L_blink'	229	1127	31	259	0	0	0	0	0
10	'L_fixation'	284	707	685	968	0	0	0	0	0
11	'41 '	838	4	0	0	0	0	0	0	0
12	'102 '	848	5	0	0	0	0	0	0	0
13	'L_saccade'	969	285	26	994	9.5000	624.3000	517.3000	416.9000	325.7000
14	'L_fixation'	995	708	66	1060	0	0	0	0	0
15	'L_saccade'	1061	286	10	1070	1.0500	639.7000	547.2000	624.6000	519.4000
16	'L_fixation'	1071	709	720	1790	0	0	0	0	0
17	'41 '	1708	6	0	0	0	0	0	0	0
18	'103 '	1709	7	0	0	0	0	0	0	0
19	'L_saccade'	1791	287	34	1824	13.8600	463.5000	178.2000	636.5000	551

Figure1. Part of the structure sEEg.event

UR-EVENTS

A separate event structure, s*EEG.urevent*, holds all the event information originally loaded into the dataset.

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EYE CHANNELS:

Added as 130-133 “electrodes”,

Channel 130 - time

Channel 131 - x coordinate of the eye

Channel 132- y coordinate of the eye

Channnel 133 - pupil size